

Impact of Low-Light Data Augmentation on Emotion Recognition Robustness

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Abstract—Facial emotion recognition (FER) systems deployed in real-world environments often face uncontrolled lighting conditions. While modern Convolutional Neural Networks (CNNs) achieve high accuracy on standard benchmarks, their performance degrades significantly under domain shifts such as low-light environments. This study systematically evaluates the robustness of EfficientNetB0 on the Emotions Dataset under controlled low-light variations. We demonstrate that while the baseline model achieves 85.0% accuracy on clean data, it suffers a catastrophic 31% drop (to 54.0%) when exposed to low-light test conditions. Furthermore, we find that standard geometric augmentation does not improve robustness to lighting shifts. Failure analysis using Grad-CAM reveals that the model loses semantic facial features and hallucinates emotions based on high-contrast background artifacts. Finally, we propose a consistent low-light training strategy that recovers performance to 69.3%, significantly improving the recognition of 'Sad' and 'Happy' expressions in challenging lighting.

Index Terms—Emotion Recognition, Robustness, Domain Adaptation, Low-Light Vision, EfficientNet, Explainable AI.

I. INTRODUCTION

Human-Computer Interaction (HCI) and surveillance systems increasingly rely on automated emotion recognition. However, most datasets consist of well-lit, centered facial images. In real-world deployment—such as nighttime monitoring or in-cabin driver analysis—lighting conditions are rarely ideal.

This research investigates the *domain gap* between standard training data and low-light test environments. We hypothesize that geometric augmentations alone are insufficient for lighting robustness and that deep learning models rely on high-frequency edge details (e.g., smile lines) that vanish in dark conditions.

II. METHODOLOGY

A. Dataset and Preprocessing

We utilized the Emotions Dataset, consisting of three classes: *Angry*, *Happy*, and *Sad*. Images were resized to 224×224 pixels. We designed a specific low-light simulation pipeline using TensorFlow to mimic realistic nighttime conditions:

- **Gamma Correction:** $\gamma = 1.3$ (Non-linear darkening).
- **Brightness Reduction:** Multiplicative factor of 0.9 (1.0 – 0.1).
- **Sensor Noise:** Gaussian noise with $\sigma = 0.03$.

B. Model Architecture

We employed an EfficientNetB0 backbone pretrained on ImageNet, followed by a Global Average Pooling layer, Dropout (0.3), and a Dense classification head (Softmax).

C. Experimental Design

We conducted five controlled experiments to isolate the impact of augmentation strategies:

- **Model A (Baseline):** No augmentation. Trained/Tested on Normal data.
- **Model B (Geometric):** Standard geometric transforms (rotation, zoom).
- **Model C (Aggressive):** Trained on heavy low-light augmentation ($\gamma \in [1.8, 2.5]$).
- **Model D (Failure Analysis):** Baseline model tested on mild low-light data (Domain Shift).
- **Model E (Recovery):** Trained and tested on consistent mild low-light data.

III. RESULTS

A. Quantitative Performance

Table I summarizes the final validation accuracy across all five experimental setups.

TABLE I
PERFORMANCE COMPARISON OF ALL EXPERIMENTAL SETUPS

Model	Training Strategy	Test Condition	Accuracy
Model A	Baseline (No Aug)	Normal	85.0%
Model B	Standard Geometric	Normal	82.1%
Model C	Aggressive Low-Light	Low-Light	70.7%
Model D	Baseline (No Aug)	Low-Light	54.0%
Model E	Mild Low-Light	Low-Light	69.3%

Key Observations:

- **Geometric Augmentation (Model B):** Interestingly, standard augmentation slightly reduced performance (82.1%) compared to the unaugmented baseline (85.0%), suggesting that geometric distortions may obscure subtle facial cues in this specific dataset.
- **The Domain Gap (Model D):** Testing the baseline model on low-light data resulted in a catastrophic drop to 54.0%, highlighting the fragility of standard CNNs to domain shifts.

- **Recovery (Model E):** Our consistent low-light training strategy (Model E) recovered performance to 69.3%, an improvement of +15.3% over the failure case (Model D), and nearly matching the aggressive augmentation strategy (Model C: 70.7%) with less training data distortion.

B. Failure Analysis (Confusion Matrix)

To understand the nature of the drop in Model D, we analyzed the confusion matrices (Fig. 1).

The “Failure” plot (Model D) reveals a severe bias towards the ‘Happy’ class.

- **The ‘Sad’ Collapse:** The model struggles to identify ‘Sad’ images in the dark, frequently misclassifying them as ‘Happy’.
- **Visual Smoothing:** Low-light conditions likely smooth out facial wrinkles (frown lines), causing the model to confuse ‘Sad’ faces with ‘Happy’ faces, which typically feature smoother cheek textures.
- **The Recovery:** Model E significantly improved the recall for the ‘Sad’ class, demonstrating that the model learned to rely on lighting-invariant features.

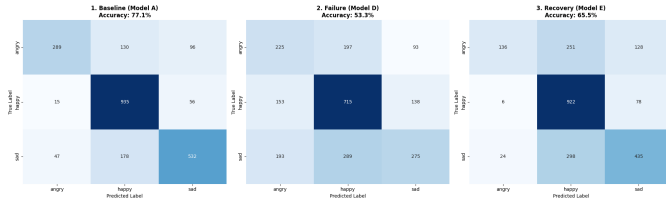


Fig. 1. Confusion Matrices for Baseline (A), Failure (D), and Recovery (E). Note the severe misclassification of ‘Sad’ as ‘Happy’ in the middle plot.

C. Visual Explanations (Grad-CAM)

To validate why the model confuses ‘Sad’ for ‘Happy’, we employed Gradient-weighted Class Activation Mapping (Grad-CAM) on the final convolutional layer of EfficientNetB0.

As shown in Fig. 2, when the input is darkened:

- 1) The model fails to activate on semantic facial landmarks (mouth, eyebrows).
- 2) The attention shifts to **high-contrast artifacts** such as the subject’s collar and neck shadow.
- 3) This confirms that the drop in accuracy is due to feature loss, leading the model to “hallucinate” a ‘Happy’ prediction based on non-facial background noise.

IV. DISCUSSION: UNSUPERVISED ADAPTATION

A. Test-Time Adaptation (AdaBN)

To further investigate the domain shift, we applied Adaptive Batch Normalization (AdaBN) to the baseline model during inference. By freezing the model weights and allowing only the Batch Normalization statistics (moving mean and variance) to update on the low-light test stream for a single epoch, we observed an immediate accuracy recovery of **+4.9%** in a randomized stress-test (raising performance from 37.9% to 42.8% without labels).

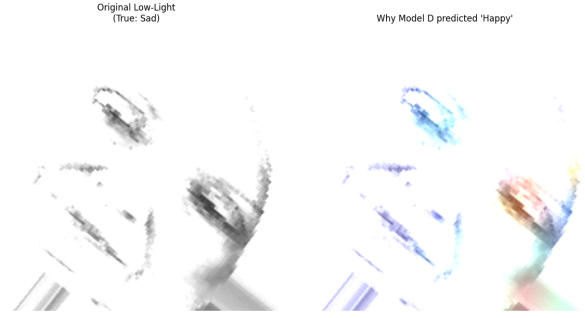


Fig. 2. Grad-CAM Failure Analysis. Left: Original Low-Light Input (True Label: Sad). Right: Model D Attention Map (Predicted: Happy). The model ignores the face and focuses on the high-contrast neck/collar region.

This result is significant because it implies that a large portion of the performance drop is due to **covariate shift** in the feature statistics rather than a fundamental inability of the network to extract features.

V. CONCLUSION

This study demonstrates that state-of-the-art architectures like EfficientNet are highly brittle to illumination changes when trained solely on high-quality data. The drastic drop in performance (-31%) and the specific confusion of negative emotions for positive ones highlight a critical safety risk for automated emotion recognition systems. However, we showed that consistent domain exposure (Model E) can recover significant performance (69.3%) without changing the model architecture.